POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5024B | Semester V | Theory | 4 Credits

Power Plant Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5024B as Power Plant Engineering in Semester V for Mechanical Engineering. The official SITTTR detailed source was unavailable. With user approval, this handbook uses reliable public diploma and industry course sources and does not represent recovered official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. Plant operation, maintenance, environmental compliance and nuclear/radiation matters require licensed/authorised procedures and competent supervision.

Verified source note: Sources support power economy, coal thermal plant systems, diesel/gas/cogeneration, nuclear and hydro plant fundamentals and industrial context.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Power Plant Engineering studies conversion of energy into dependable electric power through thermal, diesel/gas, nuclear and hydro plants, with attention to plant economics, components, operation, maintenance and environmental responsibility.

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□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□
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Prerequisite foundation

Thermodynamics, fluid mechanics, heat transfer, basic electrical concepts and strict industrial-safety awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain plant types, energy conversion and power-economy terms.
2. Explain modern coal-based thermal plant layout and major systems.
3. Explain diesel, gas turbine, cogeneration and combined-cycle plant concepts.
4. Explain nuclear and hydro plant principles, components, safety and environmental considerations.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO ▯▯▯▯▯▯ exam-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯. Define, explain, apply ▯▯▯▯▯ action words ▯▯▯▯▯▯▯▯▯▯▯▯▯.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain power-plant types, load/economy terms and energy-conversion paths.	Understand
CO2	Explain thermal-plant cycles, layout and auxiliary systems.	Understand
CO3	Compare diesel, gas-turbine, cogeneration and combined-cycle plants.	Understand
CO4	Explain nuclear and hydro components, operating principles and safety/environment constraints.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Power Generation and Plant Economy	Energy scenario, generation pathways, load curves, load factor, capacity, tariffs, costs and grid awareness.	Approx. 14 hours	CO1	Module 1
2	Coal-Based Thermal Power Plants	Rankine-cycle concept, reheating/regeneration, boilers, fuel/ash, air-gas, steam-water, condenser/cooling and emissions systems.	Approx. 16 hours	CO2	Module 2
3	Diesel, Gas-Turbine and Combined Plants	Diesel/gas plant layout, auxiliary systems, operational comparison, cogeneration and combined cycle.	Approx. 15 hours	CO3	Module 3
4	Nuclear and Hydro Power Plants	Fission/reactor components and waste awareness; hydro layout, site/resource considerations and environmental responsibility.	Approx. 15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Power Generation and Plant Economy

Energy scenario, generation pathways, load curves, load factor, capacity, tariffs, costs and grid awareness.

CO1 · Approx. 14 hours

Coal-Based Thermal Power Plants

Rankine-cycle concept, reheating/regeneration, boilers, fuel/ash, air-gas, steam-water, condenser/cooling and emissions systems.

CO2 · Approx. 16 hours

Diesel, Gas-Turbine and Combined Plants

Diesel/gas plant layout, auxiliary systems, operational comparison, cogeneration and combined cycle.

CO3 · Approx. 15 hours

Nuclear and Hydro Power Plants

Fission/reactor components and waste awareness; hydro layout, site/resource considerations and environmental responsibility.

CO4 · Approx. 15 hours

MODULE 1

Power Generation and Plant Economy

Energy scenario, generation pathways, load curves, load factor, capacity, tariffs, costs and grid awareness.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

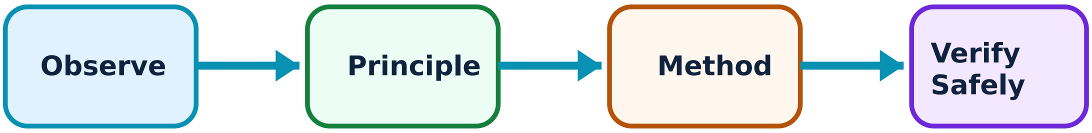
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Power Generation and Plant Economy: identify the system, state its principle, follow the correct method and verify the result safely.

1. Energy conversion and plant types−

Power plants convert chemical, thermal, nuclear, hydraulic or other energy into mechanical and electrical output. Plant choice depends on resource, demand, reliability, cost, environment and grid role.

Study and application method

Trace source energy through conversion stages to generator/grid and identify major losses/constraints.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace source energy through conversion stages to generator/grid and identify major losses/constraints.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not compare plant efficiency without a common boundary, fuel basis and operating condition.

2. Load and economics−

Load curve, load factor, diversity, fixed/operating cost and tariff concepts support capacity planning and cost assessment.

Study and application method

State time period, demand data, units and formula assumptions before computing a simple factor or cost measure.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State time period, demand data, units and formula assumptions before computing a simple factor or cost measure.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Plant-economy calculations are model-based; fuel, financing, reliability and environmental costs can change conclusions.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Coal-Based Thermal Power Plants

Rankine-cycle concept, reheating/regeneration, boilers, fuel/ash, air-gas, steam-water, condenser/cooling and emissions systems.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

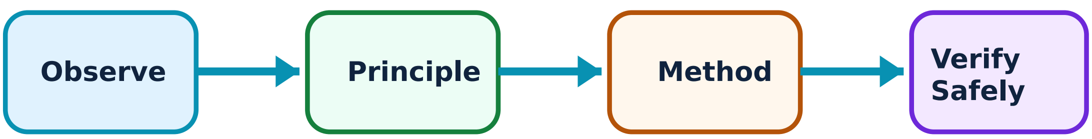
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Coal-Based Thermal Power Plants: identify the system, state its principle, follow the correct method and verify the result safely.

1. Thermal cycle and layout−

A thermal plant uses boiler heat to produce steam that expands through a turbine and is condensed/returned as feedwater. Reheat/regeneration can improve performance.

Study and application method

Draw conceptual fuel-air-boiler-turbine-condenser-feedwater flow and label energy transfers.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw conceptual fuel-air-boiler-turbine-condenser-feedwater flow and label energy transfers.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High-pressure steam systems are hazardous; never infer operating steps from a classroom layout.

2. Boilers and auxiliaries−

Boiler, furnace, feedwater, draught, fuel handling, ash handling, cooling and emission-control systems work as an integrated balance of plant.

Study and application method

For each auxiliary, state input, output, function and a likely operational/safety concern.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: For each auxiliary, state input, output, function and a likely operational/safety concern.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Boiler maintenance and combustion controls require approved permits, isolation and trained personnel.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Diesel, Gas-Turbine and Combined Plants

Diesel/gas plant layout, auxiliary systems, operational comparison, cogeneration and combined cycle.

Approx. 15 hours | CO3 | Understand · Apply

Learning goals

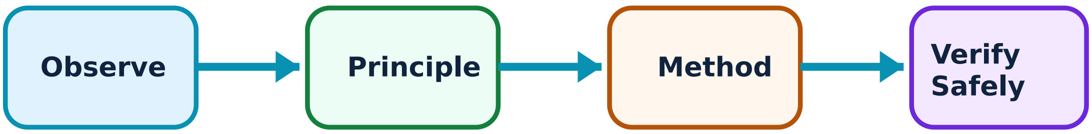
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Diesel, Gas-Turbine and Combined Plants: identify the system, state its principle, follow the correct method and verify the result safely.

1. Diesel and gas-turbine plants−

Diesel plants use reciprocating engines; gas turbines use compressor-combustor-turbine flow. They differ in start response, fuel, maintenance, efficiency and role.

Study and application method

Compare layout, conversion steps, merits/limits and suitable demand scenario in a table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare layout, conversion steps, merits/limits and suitable demand scenario in a table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Fuel, fire, noise and hot-surface hazards require engineered controls and site procedures.

2. Cogeneration and combined cycle−

Cogeneration supplies useful heat plus power; combined cycle recovers gas-turbine exhaust heat for a steam cycle, improving fuel utilisation under appropriate conditions.

Study and application method

Trace energy streams and identify where recovered heat performs useful work.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace energy streams and identify where recovered heat performs useful work.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Claimed efficiency must define included electrical/thermal outputs and fuel input.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Nuclear and Hydro Power Plants

Fission/reactor components and waste awareness; hydro layout, site/resource considerations and environmental responsibility.

Approx. 15 hours | CO4 | Understand · Apply

Learning goals

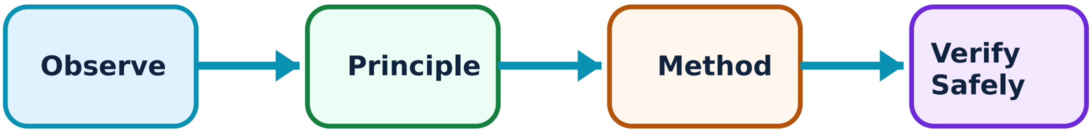
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Nuclear and Hydro Power Plants: identify the system, state its principle, follow the correct method and verify the result safely.

1. Nuclear plant fundamentals−

Controlled fission produces heat for a power cycle. Reactor, fuel, moderator, coolant, control systems and containment work within rigorous safety and regulatory frameworks.

Study and application method

Identify major functions conceptually and distinguish fission/fusion, chain reaction and containment roles.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify major functions conceptually and distinguish fission/fusion, chain reaction and containment roles.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Nuclear safety/waste topics require authoritative current sources; never generalise operating or emergency procedures.

2. Hydroelectric plants−

Hydro plants convert water head/flow into turbine-generator output. Dams, intake, penstock, turbine, generator and tailrace must be matched to site hydrology and environmental/social conditions.

Study and application method

Draw water path and list head, flow, turbine, grid and ecological data needed for assessment.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw water path and list head, flow, turbine, grid and ecological data needed for assessment.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Hydro projects require thorough geotechnical, hydrological, environmental and community evaluation.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Power Generation and Plant Economy

Use the concept flow to revise observation, principle, method and verification.

Module 2: Coal-Based Thermal Power Plants

Use the concept flow to revise observation, principle, method and verification.

Module 3: Diesel, Gas-Turbine and Combined Plants

Use the concept flow to revise observation, principle, method and verification.

Module 4: Nuclear and Hydro Power Plants

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare plant types for a stated demand/resource scenario.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a coal thermal-plant flow diagram with auxiliaries.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a simple load-factor/tariff exercise using supplied data.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a safety/environment checklist for a simulated plant study visit.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Power Generation and Plant Economy

Explain Energy conversion and plant types and state one correct method, application or precaution.−

Power plants convert chemical, thermal, nuclear, hydraulic or other energy into mechanical and electrical output. Plant choice depends on resource, demand, reliability, cost, environment and grid role.

Answer framework: Trace source energy through conversion stages to generator/grid and identify major losses/constraints.

Important point: Do not compare plant efficiency without a common boundary, fuel basis and operating condition.

Explain Load and economics and state one correct method, application or precaution.−

Load curve, load factor, diversity, fixed/operating cost and tariff concepts support capacity planning and cost assessment.

Answer framework: State time period, demand data, units and formula assumptions before computing a simple factor or cost measure.

Important point: Plant-economy calculations are model-based; fuel, financing, reliability and environmental costs can change conclusions.

Module 2 — Coal-Based Thermal Power Plants

Explain Thermal cycle and layout and state one correct method, application or precaution.−

A thermal plant uses boiler heat to produce steam that expands through a turbine and is condensed/returned as feedwater. Reheat/regeneration can improve performance.

Answer framework: Draw conceptual fuel-air-boiler-turbine-condenser-feedwater flow and label energy transfers.

Important point: High-pressure steam systems are hazardous; never infer operating steps from a classroom layout.

Explain Boilers and auxiliaries and state one correct method, application or precaution.−

Boiler, furnace, feedwater, draught, fuel handling, ash handling, cooling and emission-control systems work as an integrated balance of plant.

Answer framework: For each auxiliary, state input, output, function and a likely operational/safety concern.

Important point: Boiler maintenance and combustion controls require approved permits, isolation and trained personnel.

Module 3 — Diesel, Gas-Turbine and Combined Plants

Explain Diesel and gas-turbine plants and state one correct method, application or precaution.−

Diesel plants use reciprocating engines; gas turbines use compressor-combustor-turbine flow. They differ in start response, fuel, maintenance, efficiency and role.

Answer framework: Compare layout, conversion steps, merits/limits and suitable demand scenario in a table.

Important point: Fuel, fire, noise and hot-surface hazards require engineered controls and site procedures.

Explain Cogeneration and combined cycle and state one correct method, application or precaution.−

Cogeneration supplies useful heat plus power; combined cycle recovers gas-turbine exhaust heat for a steam cycle, improving fuel utilisation under appropriate conditions.

Answer framework: Trace energy streams and identify where recovered heat performs useful work.

Important point: Claimed efficiency must define included electrical/thermal outputs and fuel input.

Module 4 — Nuclear and Hydro Power Plants

Explain Nuclear plant fundamentals and state one correct method, application or precaution.−

Controlled fission produces heat for a power cycle. Reactor, fuel, moderator, coolant, control systems and containment work within rigorous safety and regulatory frameworks.

Answer framework: Identify major functions conceptually and distinguish fission/fusion, chain reaction and containment roles.

Important point: Nuclear safety/waste topics require authoritative current sources; never generalise operating or emergency procedures.

Explain Hydroelectric plants and state one correct method, application or precaution.−

Hydro plants convert water head/flow into turbine-generator output. Dams, intake, penstock, turbine, generator and tailrace must be matched to site hydrology and environmental/social conditions.

Answer framework: Draw water path and list head, flow, turbine, grid and ecological data needed for assessment.

Important point: Hydro projects require thorough geotechnical, hydrological, environmental and community evaluation.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Power Generation and Plant Economy.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Coal-Based Thermal Power Plants.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Diesel, Gas-Turbine and Combined Plants.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Nuclear and Hydro Power Plants.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Energy scenario, generation pathways, load curves, load factor, capacity, tariffs, costs and grid awareness..
2. Develop a complete answer or practical plan for: Rankine-cycle concept, reheating/regeneration, boilers, fuel/ash, air-gas, steam-water, condenser/cooling and emissions systems..

3. Develop a complete answer or practical plan for: Diesel/gas plant layout, auxiliary systems, operational comparison, cogeneration and combined cycle..
4. Develop a complete answer or practical plan for: Fission/reactor components and waste awareness; hydro layout, site/resource considerations and environmental responsibility..

LAST-MILE REVISION

Quick Revision

Module 1: Power Generation and Plant Economy

Energy scenario, generation pathways, load curves, load factor, capacity, tariffs, costs and grid awareness.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Coal-Based Thermal Power Plants

Rankine-cycle concept, reheating/regeneration, boilers, fuel/ash, air-gas, steam-water, condenser/cooling and emissions systems.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Diesel, Gas-Turbine and Combined Plants

Diesel/gas plant layout, auxiliary systems, operational comparison, cogeneration and combined cycle.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Nuclear and Hydro Power Plants

Fission/reactor components and waste awareness; hydro layout, site/resource considerations and environmental responsibility.

- Match each topic to CO4.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Energy conversion and plant types	Power plants convert chemical, thermal, nuclear, hydraulic or other energy into mechanical and electrical output.
Load and economics	Load curve, load factor, diversity, fixed/operating cost and tariff concepts support capacity planning and cost assessment.
Thermal cycle and layout	A thermal plant uses boiler heat to produce steam that expands through a turbine and is condensed/returned as feedwater.
Boilers and auxiliaries	Boiler, furnace, feedwater, draught, fuel handling, ash handling, cooling and emission-control systems work as an integrated balance of plant.
Diesel and gas-turbine plants	Diesel plants use reciprocating engines; gas turbines use compressor-combustor-turbine flow.
Cogeneration and combined cycle	Cogeneration supplies useful heat plus power; combined cycle recovers gas-turbine exhaust heat for a steam cycle, improving fuel utilisation under appropriate conditions.
Nuclear plant fundamentals	Controlled fission produces heat for a power cycle.
Hydroelectric plants	Hydro plants convert water head/flow into turbine-generator output.

LEARNING RESOURCES

References

Recommended power-plant-engineering references

1. P. K. Nag, Power Plant Engineering.
2. R. K. Rajput, Power Plant Engineering.
3. Domkundwar, Power Plant Engineering.

Approved public power-plant sources

- https://s3-ap-southeast-1.amazonaws.com/gtusitecirculars/Syallbus/4361905.pdf
- https://lntedutech.com/courses/power-plant-engineering-an-industrial-context/

These sources support the study handbook while official Course 5024B detail is unavailable; they do not replace authorised plant procedures.